PHYS1408 Practice Exam 2-2 — Answer Key

Worked solutions

All answers were checked numerically with `_scripts/verify_2_2.py`. $g = 9.8$ m/s^2 throughout.

This exam covers 15 problem types that are **not** on Practice Exam 2-1. Together, 2-1 and 2-2 cover 30 different question types from the professor's notes and the 2015 exam.

Q	Answer	Value	Topic (source)
1	**b**	11.7 m/s	Flat curve v_{max} (Ch 6 notes p. 2/7)
2	**c**	18.6 m/s $\rightarrow$ 20	Banked curve v_{min}, round **up** (Ch 6 notes p. 7/7, handout 2/3)
3	**b**	2.05 m/s	Atwood's machine by energy (Ch 7/8 notes p. 2/18)
4	**d**	32.8 m/s	Roller coaster (Ch 7/8 notes p. 3/18)
5	**a**	3.5 m/s	Frictionless incline, sliding up (Ch 7/8 notes p. 6/18)
6	**c**	6.1 m	Sliding down a rough incline, stopping distance (Ch 7/8 notes p. 10/18)
7	**e**	stops at 1.36 m	Incline + hanging mass, stops early (Ch 7/8 notes p. 15/18–16/18)
8	**d**	0.15 m	Mass slides into a spring (Ch 7/8 notes p. 11/18)
9	**c**	1030 W	Power climbing stairs (Ch 7/8 notes p. 14/18)
10	**e**	900 N	Force from a bounce (Ch 9 notes p. 1/5)
11	**d**	3.0 s	Rotational kinematics (Ch 10 notes p. 1/8, Ch 11 notes p. 6/8)
12	**e**	0.63 kg $\cdot$ m^2	Parallel axis theorem (Ch 10 notes p. 2/8)

Q	Answer	Value	Topic (source)
13	**b**	2.96 m/s²	Rolling acceleration $\frac{5}{7}g\sin\theta$ (Ch 10 notes p. 3/8)
14	**a**	1.05 rad/s	Person drops onto MGR at $r < R$ (Ch 11 notes p. 2/8)
15	**a**	1.36 rad/s	Rod dropped center to edge (Ch 11 notes p. 5/8)

1. Flat curve — b) 11.7 m/s

Static friction is the only horizontal force, so it supplies the centripetal force. At v_{max}, friction is at its limit, $F_f = \mu_s N$ ("max it can be!").

$$N = mg, \qquad \mu_s mg = \frac{mv^2}{r} \quad \Rightarrow \quad v_{max} = \sqrt{\mu_s g r} = \sqrt{0.40 \times 9.8 \times 35} = \sqrt{137.2} = 11.7 \text{ m/s}.$$

(N = normal force, μ_s = coefficient of static friction, r = radius of the curve.)

Distractors:

- a) 3.7 forgets g: $\sqrt{\mu_s r}$.
- c) 16.6 has an extra 2: $\sqrt{2\mu_s g r}$.
- d) 18.5 drops μ_s: $\sqrt{gr}$.
- e) 137 forgets the square root.

2. Banked curve, minimum speed — c) 20 m/s

At v_{min} the car tends to slide **down** the bank, so static friction points **up** the bank. This flips every μ sign compared with v_{max} (Ch 6 notes p. 7/7):

$$v_{min} = \sqrt{gr\,\frac{\tan\theta - \mu_s}{1 + \mu_s \tan\theta}} = \sqrt{9.8 \times 90 \times \frac{0.5774 - 0.15}{1 + 0.15 \times 0.5774}} = \sqrt{882 \times \frac{0.4274}{1.0866}} = \sqrt{346.9} = 18.6 \text{ m/s}.$$

(θ = bank angle from the horizontal.)

Answer-choice trap. 18.6 m/s is a **minimum**. A car going 18 m/s slides down the bank, so b) 18 doesn't work even though it's the closest number. The smallest choice that works is **c) 20 m/s**.

Distractors:

- a) 12 is the flat-curve formula, $\sqrt{\mu_s g r} = 11.5$.
- d) 23 is the frictionless bank, $\sqrt{gr\tan\theta} = 22.6$.
- e) 27 is $v_{max} = 26.5$.

3. Atwood's machine — b) 2.05 m/s

$E_i = E_f$ for both masses together (Ch 7/8 notes p. 2/18). m_1 starts at height h and ends at 0; m_2 starts at 0 and ends at h. Both move at the same speed v, starting from rest.

Energy terms:

$$E_i = m_1 g h$$

$$E_f = \tfrac{1}{2} m_1 v^2 + \tfrac{1}{2} m_2 v^2 + m_2 g h$$

Solve for v:

$$m_1 g h = \tfrac{1}{2}(m_1 + m_2) v^2 + m_2 g h$$

$$(m_1 - m_2) g h = \tfrac{1}{2}(m_1 + m_2) v^2$$

$$v = \sqrt{\frac{2gh(m_1 - m_2)}{m_1 + m_2}}$$

Plug in:

$$v = \sqrt{\frac{2(9.8)(1.5)(8.0 - 6.0)}{8.0 + 6.0}} = \sqrt{4.20} = 2.05 \text{ m/s}$$

Distractors:

- a) 1.45 forgets the 2.
- c) 2.71 divides by m_1 only.
- d) 3.13 divides by m_2 only.
- e) 5.42 is free fall, $\sqrt{2gh}$.

4. Roller coaster — d) 32.8 m/s

No friction, so $E_1 = E_n$ (Ch 7/8 notes p. 3/18). The hump in between doesn't matter, only the start and end heights.

Energy terms:

$$E_1 = m g h_1$$

$$E_n = \tfrac{1}{2} m v^2 + m g h_n$$

Solve for v:

$$mgh_1 = \tfrac{1}{2}mv^2 + mgh_n$$

$$v^2 = 2g(h_1 - h_n)$$

$$v = \sqrt{2g(h_1 - h_n)}$$

Plug in:

$$v = \sqrt{2(9.8)(65 - 10)} = \sqrt{1078} = 32.8 \text{ m/s}$$

Distractors:

- a) 14.0 uses only the final 10 m.
- b) 19.8 uses the $65 - 45$ hump drop.
- c) 23.2 forgets the 2.
- e) 35.7 uses the full 65 m drop to the ground.

5. Frictionless incline, sliding up — a) 3.5 m/s

Frictionless, so $E_i = E_f$ (Ch 7/8 notes p. 6/18). Put $h = 0$ at the start; after sliding Δx up the slope it is $\Delta x \sin\theta$ higher.

Energy terms:

$$E_i = \tfrac{1}{2}mv_i^2$$

$$E_f = \tfrac{1}{2}mv_f^2 + mg\Delta x \sin\theta$$

Solve for v_f:

$$\tfrac{1}{2}mv_i^2 = \tfrac{1}{2}mv_f^2 + mg\Delta x \sin\theta$$

$$v_f^2 = v_i^2 - 2g\Delta x \sin\theta$$

$$v_f = \sqrt{v_i^2 - 2g\Delta x \sin\theta}$$

Plug in:

$$v_f = \sqrt{5.0^2 - 2(9.8)(2.5)(0.2588)} = \sqrt{25 - 12.68} = \sqrt{12.32} = 3.5 \text{ m/s}$$

Distractors:

- b) 4.3 forgets the 2.
- c) 6.1 adds instead of subtracts.
- d) 12.3 forgets the square root.
- e) is what you get if you use $\cos\theta$, or treat 2.5 m as the height.

6. Sliding down a rough incline, stopping distance — c) 6.1 m

$E_i = E_f + E_{lost}$ (Ch 7/8 notes p. 10/18). Put $h = 0$ where the block stops, so it starts $\Delta x \sin\theta$ higher. Going down, gravity **adds** energy and friction removes it. $N = mg\cos\theta$ on the incline.

Energy terms ($v_f = 0$):

$$E_i = \tfrac{1}{2}mv_i^2 + mg\Delta x \sin\theta$$

$$E_f = 0$$

$$E_{lost} = \mu_k mg\cos\theta\,\Delta x$$

Solve for Δx:

$$\tfrac{1}{2}mv_i^2 + mg\Delta x\sin\theta = \mu_k mg\cos\theta\,\Delta x$$

$$\tfrac{1}{2}v_i^2 = g\Delta x(\mu_k\cos\theta - \sin\theta)$$

$$\Delta x = \frac{v_i^2}{2g(\mu_k\cos\theta - \sin\theta)}$$

This is the professor's form, with the bracket written so the answer is positive.

Plug in:

$$\Delta x = \frac{4.0^2}{2(9.8)(0.35\times 0.9781 - 0.2079)} = \frac{16}{19.6(0.1344)} = 6.1 \text{ m}$$

It does stop, because $\mu_k\cos\theta = 0.342$ is bigger than $\sin\theta = 0.208$. That rules out e).

Distractors:

- a) 1.5 uses the sliding-**up** bracket, $(\sin\theta + \mu_k\cos\theta)$.
- b) 2.4 drops the $\sin\theta$ term.
- d) 12.1 forgets the 2.

7. Incline + hanging mass — e) it stops before 1.5 m

$E_i = E_f + E_{lost}$ for both masses (Ch 7/8 notes p. 15/18). Both move the same distance Δx at the same speed.

- m_1 starts at 0 and rises $\Delta x\sin\theta$.
- m_2 starts at Δx and drops to 0.
- Friction acts on m_1, with $N = m_1 g\cos\theta$.

Energy terms:

$$E_i = \tfrac{1}{2}(m_1 + m_2)v_i^2 + m_2 g \Delta x$$

$$E_f = \tfrac{1}{2}(m_1 + m_2)v_f^2 + m_1 g \Delta x \sin\theta$$

$$E_{lost} = \mu_k m_1 g \cos\theta \, \Delta x$$

Solve for v_f:

$$\tfrac{1}{2}(m_1 + m_2)v_i^2 + m_2 g \Delta x = \tfrac{1}{2}(m_1 + m_2)v_f^2 + m_1 g \Delta x \sin\theta + \mu_k m_1 g \cos\theta \, \Delta x$$

$$v_f^2 = v_i^2 + \frac{2g\Delta x}{m_1 + m_2}\left(m_2 - m_1 \sin\theta - \mu_k m_1 \cos\theta\right)$$

$$v_f = \sqrt{v_i^2 + \frac{2g\Delta x}{m_1 + m_2}\left(m_2 - m_1 \sin\theta - \mu_k m_1 \cos\theta\right)}$$

Plug in. The bracket is $4.0 - 15(0.4226) - 0.30(15)(0.9063) = 4.0 - 6.339 - 4.078 = -6.417$ kg, so

$$v_f = \sqrt{3.0^2 + \frac{2(9.8)(1.5)}{19}(-6.417)} = \sqrt{9.0 - 9.93} = \sqrt{-0.93}$$

The number under the root is negative ("imaginary"), so it "doesn't go 1.5 m."

Where it does stop. Set $v_f = 0$ in the v_f^2 line and solve for Δx:

$$0 = v_i^2 + \frac{2g\Delta x}{m_1 + m_2}\left(m_2 - m_1 \sin\theta - \mu_k m_1 \cos\theta\right)$$

$$\Delta x = \frac{(m_1 + m_2)v_i^2}{-2g\left(m_2 - m_1 \sin\theta - \mu_k m_1 \cos\theta\right)}$$

$$\Delta x = \frac{19(9.0)}{2(9.8)(6.417)} = 1.36 \text{ m}$$

Distractors:

- a) 2.32 ignores friction.
- b) 2.51 uses the sliding-down bracket.
- c) 2.98 drops the $m_1 \sin\theta$ term.
- d) 4.35 adds the bracket instead of subtracting it.

8. Mass slides into a spring — d) 0.15 m

No friction, so $E_i = E_f$ (Ch 7/8 notes p. 11/18). Put $h = 0$ at the floor. All the gravitational PE becomes spring PE.

Energy terms (at rest at both ends):

$$E_i = mgh$$

$$E_f = \tfrac{1}{2}kx^2$$

Solve for x:

$$mgh = \tfrac{1}{2}kx^2$$

$$x^2 = \frac{2mgh}{k}$$

$$x = \sqrt{\frac{2mgh}{k}}$$

Plug in:

$$x = \sqrt{\frac{2(2.5)(9.8)(0.80)}{1800}} = \sqrt{0.02178} = 0.15 \text{ m}$$

Distractors:

- a) 0.011 is mgh/k.
- b) 0.022 forgets the square root.
- c) 0.10 forgets the 2.
- e) 0.21 doubles it.

9. Power — c) 1030 W

Work done against gravity is mgh, and power is work per time (Ch 7/8 notes p. 14/18):

$$P = \frac{mgh}{t} = \frac{70 \times 9.8 \times 12}{8.0} = 1029 \text{ W} \approx 1030 \text{ W}.$$

Distractors:

- a) 105 forgets g.
- b) 457 swaps h and t.
- d) 8230 is the work mgh in joules, not power.
- e) 65,900 multiplies by t.

10. Force from a bounce — e) 900 N

$F = \dfrac{p_f - p_i}{t}$. The ball reverses direction, so $p_i = -p_f$ and $|F| = \dfrac{2p}{t}$ ("100% rebound", Ch 9 notes p. 1/5):

$$|F| = \frac{2mv}{t} = \frac{2 \times 0.15 \times 30}{0.010} = 900 \text{ N}.$$

Distractors:

- a) 0 subtracts the speeds and ignores direction.
- b) 4.5 is the momentum $p = mv$ (kg · m/s), not a force.
- c) 9.0 is the change in momentum (N · s).
- d) 450 forgets that a bounce doubles the change in momentum.

11. Rotational kinematics — d) 3.0 s

No α is given and t is wanted, so use the average-velocity equation (Ch 10 notes p. 1/8):

$$\theta_f - \theta_i = \frac{\omega_i + \omega_f}{2} t \quad \Rightarrow \quad t = \frac{\Delta\theta}{(\omega_i + \omega_f)/2} = \frac{90}{(10+50)/2} = \frac{90}{30} = 3.0 \text{ s}.$$

Check: $\alpha = (\omega_f - \omega_i)/t = 13.3$ rad/s², and $\omega_f^2 = \omega_i^2 + 2\alpha\Delta\theta$ gives $100 + 2(13.3)(90) = 2500 = 50^2$.

Distractors (each divides 90 rad by the wrong speed):

- a) 1.5 divides by the sum, 60.
- b) 1.8 divides by ω_f.
- c) 2.25 divides by $\Delta\omega$.
- e) 9.0 divides by ω_i.

12. Parallel axis theorem — e) 0.63 kg · m²

Parallel axis theorem (Ch 10 notes p. 2/8): $I = I_{cm} + md^2$, where d **is measured from the rod's CENTER**, not from its end. (Worked guide 10.2, Example B.)

Step 1: I_{cm} for a rod.

$$I_{cm} = \tfrac{1}{12} m\ell^2 = \tfrac{1}{12}(3.0)(1.2)^2 = 0.36 \text{ kg} \cdot \text{m}^2$$

Step 2: d from the center. The center is $\ell/2 = 0.60$ m from the end; the axis is $a = 0.30$ m from that end:

$$d = \frac{\ell}{2} - a = 0.60 - 0.30 = 0.30 \text{ m}$$

Step 3: Add md^2.

$$I = I_{cm} + md^2 = 0.36 + (3.0)(0.30)^2 = 0.36 + 0.27 = 0.63 \text{ kg} \cdot \text{m}^2$$

Notecard shortcut:

$$I = m\left[\frac{\ell^2}{12} + \left(\frac{\ell}{2} - a\right)^2\right]$$

$$= 3.0\,(0.12 + 0.09) = 0.63 \text{ kg} \cdot \text{m}^2$$

Distractors:

- a) 0.09 is $\frac{1}{3}md^2$.
- b) 0.27 is md^2 alone.
- c) 0.36 is I_{cm} alone (forgot to move the axis).
- d) 0.50 uses $\frac{1}{2}md^2$.

13. Rolling acceleration — b) 2.96 m/s²

From forces and torque (Ch 10 notes p. 3/8):

- Along the slope: $mg\sin\theta - f = ma$.
- Torque about the center: $Rf = \frac{2}{5}mR^2\alpha$ with $R\alpha = a$, so $f = \frac{2}{5}ma$.

Combining them, $g\sin\theta = \frac{7}{5}a$:

$$a = \tfrac{5}{7}g\sin\theta = \tfrac{5}{7} \times 9.8 \times \sin 25^\circ = \tfrac{5}{7} \times 4.142 = 2.96 \text{ m/s}^2.$$

Distractors:

- a) 2.07 is a ring, $\frac{1}{2}g\sin\theta$.
- c) 4.14 is sliding, $g\sin\theta$ ("less than sliding").
- d) 6.34 uses $\cos\theta$.
- e) 7.00 forgets $\sin\theta$.

14. Person drops onto a merry-go-round — a) 1.05 rad/s

$L_i = L_f$. The person is a point mass at $r = 1.5$ m, so by the parallel axis theorem $I = 0 + mr^2$ (Ch 11 notes p. 2/8). The person had no angular momentum before stepping on.

$$\omega_f = \frac{\frac{1}{2}MR^2\omega_i}{\frac{1}{2}MR^2 + mr^2} = \frac{937.5 \times 1.2}{937.5 + 60(1.5)^2} = \frac{1125}{1072.5} = 1.05 \text{ rad/s}.$$

Distractors:

- b) 1.10 uses mr instead of mr^2.
- c) 1.12 treats the person as a disk, $\frac{1}{2}mr^2$.
- d) 1.20 is no change.
- e) 1.37 flips the ratio (as if the person stepped **off**).

15. Rod dropped from the center to the edge — a) 1.36 rad/s

The rod lies along a radius, so it rotates about **one end**: $I_{rod} = \frac{1}{3}m\ell^2$ (Ch 11 notes p. 5/8).

$$\omega_f = \frac{\frac{1}{2}MR^2\omega_i}{\frac{1}{2}MR^2 + \frac{1}{3}m\ell^2} = \frac{1800 \times 1.5}{1800 + \frac{1}{3}(60)(3.0)^2} = \frac{2700}{1980} = 1.36 \text{ rad/s}.$$

Distractors:

- b) 1.43 forgets the $\frac{1}{2}$ on the disk.
- c) 1.46 uses the rod-about-its-middle formula, $\frac{1}{12}m\ell^2$.
- d) 1.50 ignores the rod.
- e) 1.73 flips a mass ratio, $\omega_i(M+m)/M$.